

VLMs Evaluated

Name	Provider	Size	Weights
Pixtral-Large-2411	MistralAI	123B	Open
Qwen2.5-vl-72b-instruct	Qwen	72B	Open
Llama-3.2-90B-Vision-Instruct	Meta	90B	Open
Llama-3.2-11B-Vision-Instruct	Meta	11B	Open
Grok-2-Vision-1212	xAI	Big	Closed
Gemini 2.0 Flash	Google	Small	Closed
Claude 3.5 Sonnet	Anthropic	Big	Closed
GPT-4o	OpenAI	Big	Closed

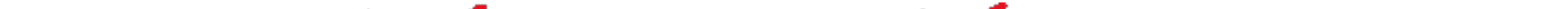
💰 To reduce the evaluation costs, we selected only the best performing VLMs for the harder challenges:

- Sanity Check (8 VLMs)

- Static Evaluations (4 VLMs)

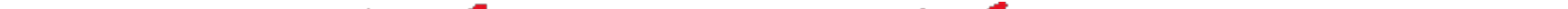
- Interactive Evaluation (2 VLMs)













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Prompts design

System prompt

User prompt

Assistant Answer

The prompts are always composed by:

- A **System message**
  - **Role** (e.g. "…physics expert analyzing…)
  - **Simulation parameters** (optional)
  - **Task description**
  - **Response format**
- **Few-shot examples** (optional) each composed by:
  - **A User message** with the example (randomly sampled)
  - **An Assistant message** with the correct response
- **A User message**:
  - With the actual question



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Simulation conditions for all tasks:
- Gravity: 9.81m/s²(downward)
- Objects density: 0.25
- Friction coefficient: 0.5
- Elasticity coefficient: 0.2
- Simulationduration:25 seconds or until objects stop moving
- Black and purple objects: static (fixed)
- Green, red, grey objects: dynamic (can move)
- Goal criterion: Objects must remain in contact ≥ 3 seconds
- Objects cannot leave the visible simulation boundaries

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Clearly define your solution by specifying:
- "x": horizontal position of the ball center (0 is left, maximum is 256 on the right)
- "y": vertical position of the ball center (0 is bottom, maximum is 256 at the top)
- "radius": size of the ball (minimum 2,maximum 32)

Important rules for placing the ball:
- The ball must remain fully within the visible simulation boundaries.
- The ball cannot overlap with existing objects.

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